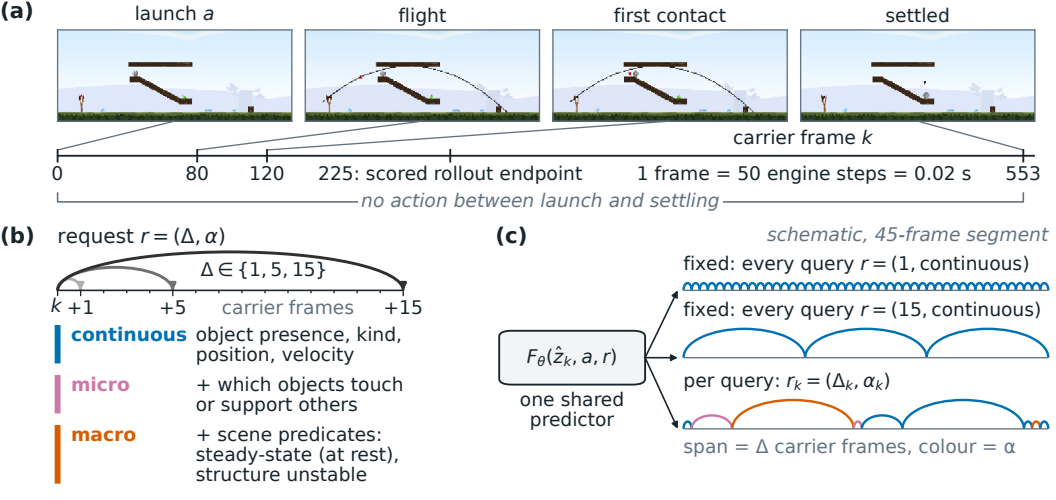

(a)
launch a
flight
first contact
settled
carrier frame k
0
80
120
225: scored rollout endpoint
1 frame = 50 engine steps = 0.02 s
553
no action between launch and settling
(b)
request r = (Δ, α)
Δ ∈ {1, 5, 15}
k +1
+5
carrier frames
+15
continuous
object presence, kind, position, velocity
micro
+ which objects touch or support others
macro
+ scene predicates: steady-state (at rest), structure unstable
(c)
schematic, 45-frame segment
fixed: every query r = (1, continuous)
fixed: every query r = (15, continuous)
per query: r_k = (Δ_k, α_k)
F_θ(ẑ_k, a, r)
one shared predictor
span = Δ carrier frames, colour = α